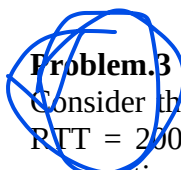


Problem Set-3


$$\text{throughput} = 65535 \cdot 8 / 2 \cdot 10^6 \quad \text{eff} = \text{throughput} / \text{bandwidth}$$

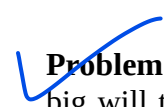
Problem.1 A TCP machine is sending full windows of 65,535 bytes over a 1-Gbps channel that has a 10-msec one-way delay. What is the maximum throughput achievable? What is the line efficiency?

Problem.2 If the TCP round-trip time, RTT, is currently 30 msec and the following acknowledgements come in after 26, 32, and 24 msec, respectively, what is the new RTT estimate using the Jacobson algorithm? Use $\alpha = 0.1$.



Problem.3 What will happen when we double the size of window in TCP congestion control? Consider the case when connection begins and having **CongWin**, $W = 1$ MSS of 500 bytes & RTT = 200 msec initial rate = 20 kbps. Find out the average throughput for only these two consecutive TCP connections, first having window size W and next one $2W$.

$$w / \text{rtt} \text{ is throughput}$$



Problem.4 Suppose that the TCP congestion window is set to 18 KB and a timeout occurs. How big will the window be if the next four transmission bursts are all successful? Assume that the maximum segment size is 1 KB. 2^4



Problem.5 Consider sending an object of size $O = 100$ Kbytes from server to client. Let $S = 536$ bytes and $\text{RTT} = 100$ msec. Suppose the transport protocol uses static windows with window size W .

a) For a transmission rate of 28 Kbps, determine the minimum possible latency. Determine the minimum window size that achieves this latency.

$$W \geq R \cdot \text{RTT} / S$$

b) Repeat (a) for 100 Kbps.

c) Repeat (a) for 1 Mbps.

d) Repeat (a) for 10 Mbps.